

Paper Information Record — Santamaria2024

Paper ID: 0022 | Oxford Missing Data Study

General Information

Field	Value
Paper ID	0022
Title	Demand-Pull Innovation and Startup Performance (SMJ 3560)
Authors	Santamaria, S., Abolfathi, N., & Mahmood, I. P.

Data Structure

Field	Value
Data type	Panel RCT (startup-level, pre/post rounds)
Observations (N)	472 total; 394 regression sample (after FE absorption)
Preferred model column	Table 3, Panel A (DiD)

Core Regression Model

Field	Value
Dependent variable	RevenueLikert (Likert 1-5, treated as continuous)
Focal IV	Post_x_Treatment (Post × Treatment DiD interaction)
Controls	Post, Age, WorkExp, EntrepExperience (augmented for simulation)
Fixed effects	SN (entity FE, startup level)
Clustered SEs	None (robust/hetero SEs)

Simulation Configuration

Field	Value
Key variables (3)	Age, WorkExp, EntrepExperience
MAR control	TeamSize
Mechanisms	MCAR, MAR, NMAR
Missingness proportions	1%, 5%, 10%, 20%, 30%, 40%, 50%
Methods	LD, Mean, Reg, Iter, RF, DL, MILGBM
Iterations per scenario	30
MAR/NMAR strength	1.5
M (multiple imputation)	5

Baseline Validation

Field	Value
Baseline coefficient	0.3739 (Post_x_Treatment)
Baseline SE	0.1913

Field	Value
Baseline p-value	0.052
Published value	~0.37 (borderline significant DiD, Table 3 Panel A)
R-squared	0.708

Run Log

Field	Value
Full run start	2026-04-10
Full run complete	2026-04-10
Total combinations	13230 (3x7x3x7x30)
Errors	0
DL method	TensorFlow 2.21.0, MLP 32->Dropout(0.1)->16->1, ReLU, Adam lr=0.005, EarlyStopping
MILGBM method	LightGBM 4.6.0, MICE M=5, Rubin's Rules, 30 estimators, max_depth=4
Note	Augmented spec: Age/WorkExp/EntrepExperience added as controls to enable key-var missingness simulation